Customizing Excel settings allows you to personalize the interface and behavior of the application to match your specific engineering or data analysis workflow. Most of these adjustments are made within the **Excel Options** menu (File > Options).

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1. General Interface Customization

These settings change the visual look and basic behavior of Excel when you first open a file.

- **User Interface Options**: You can change the "Office Theme" to **Dark Gray** or **Black** to reduce eye strain during long coding or data entry sessions.
- **Default Font and View**: Set the default font size and the initial view (e.g., **Page Break Preview**) for all new workbooks you create.
- **Startup Options**: Choose whether you want the Start screen to appear every time you launch Excel.

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2. Formula and Calculation Settings

This is a critical area for ensuring your mathematical models—like those used in **Neural Networks** or **Deep Learning** architectures—behave predictably.

- **Calculation Options**: Switch between **Automatic** (calculates every time a cell changes) and **Manual** (calculates only when you press F9). Manual calculation is a "hack" to speed up Excel when working with massive datasets that fill all **1,048,576 rows**.
- **Error Checking**: Customize which rules trigger the "green triangle" error indicator, such as "Formulas referring to empty cells" or "Inconsistent formulas in a table."
- **Working with Formulas**: Enable or disable "Autocomplete" to help you find functions like **INDEX**, **MATCH**, or **SUMIF** faster as you type.

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3. Data and Advanced Editing

Refine how Excel handles the information you **Import** or type manually.

- **Decimal Places**: Automatically fix the number of decimal places for all entries, which is useful for maintaining consistency in financial or scientific data.
- **Autocomplete for Cell Values**: Toggle this to prevent Excel from automatically filling in text based on previous entries, which helps avoid errors in columns like "Student Name."
- **Image Size and Quality**: Adjust the resolution of images or **Charts** you insert into your sheets to manage the overall file size.

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4. Customizing the Ribbon and Quick Access Toolbar

You can streamline your access to the tools you use most frequently.

- **Quick Access Toolbar (QAT)**: Add specific buttons (like **Conditional Formatting**, **Pivot Tables**, or **Protect Sheet**) to the very top of your Excel window so they are always one click away, regardless of which tab you are on.
- **Customize Ribbon**: Create your own custom tab (e.g., a "Data Science" tab) and group your favorite tools together, such as **Data Validation**, **What-If Analysis**, and **Formula Auditing**.
- **Developer Tab**: By default, this tab is hidden. You must enable it in the Ribbon settings to access **Macros** and **VBA** for advanced automation tasks.

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5. Proofing and AutoCorrect

Excel can help you maintain professional standards in your reports and documentation.

- **AutoCorrect Options**: Create your own shorthand (e.g., typing "BE-AIDS" to automatically expand to "Bachelor of Engineering in Artificial Intelligence and Data Science").
- **Dictionary Settings**: Change the language for spell-checking or add technical engineering terms to your custom dictionary so they aren't flagged as errors.

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6. Saving and Recovery

Protect your work from unexpected crashes or system errors.

- **AutoRecover**: Adjust the frequency (e.g., every 5 minutes) that Excel saves a temporary copy of your work.
- **Default File Location**: Set the specific folder on your computer where Excel should automatically save your files.